Jaypee Institute of Information Technology, Noida

Department of Computer Applications

PROJECT-BASED LEARNING REPORT

COMPENSATION ANALYTICS & PAY EQUITY:

Regression-Based Salary Analysis and Excel-Based Salary Calculator

Submitted by:

Suhani Sushilkumar Singh 992509260001
Bhavninder Singh Virk 992509260003

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Submitted to: Prof. Praveen Sharma

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ABSTRACT

Compensation analytics has emerged as a critical function within modern Human Resource Management, enabling organisations to make data-driven decisions regarding pay structures, employee retention, and pay equity. This project presents a comprehensive analysis of compensation patterns in the global data science and technology sector, leveraging two publicly available secondary datasets: the Kaggle Data Science Salaries dataset (2020–2022) comprising 607 records and the Stack Overflow Developer Survey 2024 comprising 1,926 filtered respondents.

The study encompasses a full analytical pipeline, beginning with data collection and preprocessing, followed by Exploratory Data Analysis (EDA), statistical hypothesis testing, correlation analysis, and an interactive Excel-based analytical toolkit. The toolkit incorporates pivot tables, professional charts, an interactive dashboard with dynamic slicers, and a salary calculator employing the AVERAGEIFS formula to provide personalised salary estimates.

Key findings indicate that experience level is the most significant determinant of salary, with senior professionals earning approximately 40–55% more than entry-level counterparts. Company size and geographic region also exhibit meaningful associations with compensation. Remote work arrangements demonstrate a moderate and nuanced relationship with salary outcomes.

Keywords: *Compensation Analytics, Pay Equity, HR Analytics, Data Science Salaries, Salary Calculator, Excel Analytics, Exploratory Data Analysis, AVERAGEIFS.*

1. INTRODUCTION

1.1 Overview of Compensation Analytics

Compensation analytics refers to the systematic use of quantitative methods, statistical analysis, and data visualisation to understand, evaluate, and optimise how organisations compensate their employees. In an era characterised by talent scarcity and heightened awareness of workplace equity, compensation analytics has evolved from a peripheral HR function to a strategic organisational capability. By analysing salary structures, pay bands, and compensation benchmarks, organisations can attract and retain talent more effectively, reduce turnover costs, and ensure adherence to equitable compensation principles.

The data science and technology sector presents a particularly compelling domain for compensation analytics due to its rapid growth, acute talent competition, and global workforce distribution. Salary disparities in this sector are well-documented and influenced by a multitude of factors including professional experience, employer size, geographic location, and the mode of employment.

1.2 Importance of Pay Equity

Pay equity refers to the principle that employees performing work of equal value should receive equal compensation, irrespective of gender, ethnicity, age, or other protected characteristics. Beyond its ethical imperative, pay equity has emerged as a regulatory and reputational concern for organisations globally. Unaddressed pay disparities can result in legal liability, diminished employee morale, higher attrition rates, and reputational damage. Analytical approaches to pay equity involve not only descriptive comparisons of salary distributions but also regression-based techniques to control for legitimate pay determinants such as experience and job function.

1.3 Objectives of the Study

- To analyse compensation patterns in the global data science and technology sector using real-world survey data.
- To identify key determinants of salary variation, including experience level, company size, remote work, and geographic region.
- To perform statistical analysis including hypothesis tests and correlation examination to quantify associations with compensation.
- To develop an interactive Excel-based analytical toolkit: pivot tables, charts, a dashboard, and a salary calculator.
- To provide actionable insights for HR professionals regarding pay equity and compensation strategy.

1.4 Scope and Limitations

This project analyses secondary data from two public sources and does not involve primary data collection. The scope is limited to data science and technology professionals. The analysis covers descriptive statistics, EDA, and Excel-based tools; machine learning modelling and real-time data integration are explicitly excluded from this project phase.

2. RESEARCH DESIGN & PROBLEM STATEMENT

2.1 Problem Statement

Despite the proliferation of salary data in the public domain, there remains a gap in accessible, structured analytical frameworks that HR practitioners can use to derive actionable compensation insights without requiring advanced programming skills. This project addresses this gap by combining Python-based EDA with Excel-based interactive tools, creating a dual-layer analytical approach that is both rigorous and practically accessible.

2.2 Research Questions

- RQ1: How do salary levels differ across experience levels (Entry, Mid, Senior, Executive) in the data science and technology sector?
- RQ2: What is the relationship between company size (Small, Medium, Large) and employee compensation?
- RQ3: To what extent does remote work mode (on-site, hybrid, fully remote) influence salary outcomes?
- RQ4: Are there significant salary disparities across geographic regions?
- RQ5: Can a simple Excel-based salary calculator provide reasonably accurate personalised salary estimates?

2.3 Research Methodology Overview

This project adopts a quantitative, descriptive research design structured in five stages: (1) Data Collection — secondary data acquisition from publicly available sources; (2) Data Preprocessing — cleaning, transformation, and feature engineering in Python; (3) Exploratory Data Analysis — visualisation and statistical exploration; (4) Statistical Analysis — hypothesis testing and correlation analysis; (5) Excel-Based Analysis — pivot tables, charts, interactive dashboard, and salary calculator.

3. DATA COLLECTION & METHODOLOGY

3.1 Data Sources

3.1.1 Primary Dataset: Kaggle Data Science Salaries

The first dataset, sourced from Kaggle (Bhatia, 2022) [1], contains 607 records of data science job salaries spanning 2020 to 2022. This curated dataset provides standardised salary information across 50+ countries with detailed categorisations of experience level, company size, and remote work arrangements. All salaries are normalised to USD for consistent cross-national comparison.

3.1.2 Secondary Dataset: Stack Overflow Developer Survey 2024

The second dataset comprises responses from the 2024 Stack Overflow Developer Survey [2], one of the largest surveys of software professionals worldwide. From the total 49,191 responses, a filtered subset of 1,926 data professionals with complete compensation information was extracted for analysis.

3.1.3 Rationale for Dual Dataset Approach

The combination of these two datasets provides both depth and breadth for robust compensation analysis. The Kaggle dataset offers granular role-level data with consistent USD normalisation, while the Stack Overflow survey provides a broader, more demographically diverse sample. Triangulating findings across both datasets enhances the reliability and validity of conclusions.

3.2 Data Variables

Variable	Type	Dataset	Description
experience_level	Categorical	Dataset 1	EN/MI/SE/EX (Entry/Mid/Senior/Executive)
salary_in_usd	Continuous	Dataset 1	Normalised annual salary in USD
company_size	Categorical	Dataset 1	S/M/L (Small/Medium/Large)
remote_ratio	Ordinal	Dataset 1	0, 50, or 100 (% remote work)
job_title	Categorical	Dataset 1	Specific job designation
work_year	Ordinal	Dataset 1	Year of employment (2020–2022)
ConvertedCompYearly	Continuous	Dataset 2	Annual compensation in USD
OrgSize	Categorical	Dataset 2	Organisation size category
RemoteWork	Categorical	Dataset 2	Remote/Hybrid/In-person
Country	Categorical	Dataset 2	Respondent country

4. DATA PREPROCESSING

4.1 Preprocessing Steps

Data preprocessing was conducted in Python using the pandas and NumPy libraries in a Google Colab environment. The following steps were applied to both datasets:

4.1.1 Handling Missing Values

Both datasets were assessed for missing values using the isnull() method. Dataset 1 exhibited no null values in key analytical columns. Dataset 2 required filtering to include only respondents who reported ConvertedCompYearly, resulting in a reduction from 49,191 to 1,926 records.

4.1.2 Removal of Duplicates

Duplicate records were identified and removed using pandas' drop_duplicates() method. Dataset 1 had 42 duplicate entries removed; Dataset 2 had no exact duplicate records after the initial filtering.

4.1.3 Outlier Treatment

Salary outliers were identified using the Interquartile Range (IQR) method. Records with salaries below Q1 − 1.5*IQR or above Q3 + 1.5*IQR were removed. This resulted in the removal of records with salaries below approximately $16,000 and above approximately $290,000 in Dataset 1.

4.1.4 Feature Engineering

Several derived features were created to facilitate analysis:

- experience_numeric: Ordinal encoding of experience levels (EN=1, MI=2, SE=3, EX=4).
- company_size_encoded: Ordinal encoding of company size (S=1, M=2, L=3).
- region: Derived from company_location by mapping ISO country codes to continental regions.
- remote_category: Labelled categorisation (0='On-site', 50='Hybrid', 100='Fully Remote').
- salary_band: Binning into discrete bands (<$50k, $50k–$100k, $100k–$150k, $150k–$200k, >$200k).

4.2 Post-Preprocessing Summary Statistics

Metric	Dataset 1 (Kaggle)	Dataset 2 (Stack Overflow)
Records After Cleaning	565	1,772
Mean Salary (USD)	$110,610	$99,177
Median Salary (USD)	$100,000	$81,210
Minimum Salary (USD)	$16,000	$1,000
Maximum Salary (USD)	$289,000	$500,000
Standard Deviation	$52,844	$94,328

5. EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was conducted using Python libraries including matplotlib, seaborn, and pandas. The objective was to understand the distributional characteristics of salary data and to identify meaningful patterns prior to formal statistical analysis.

5.1 Salary Distribution Analysis

The distribution of salaries in both datasets was examined using histograms and kernel density estimation (KDE) plots. Dataset 1 exhibits a right-skewed distribution with a concentration of salaries between $80,000 and $150,000. Dataset 2 is more heavily right-skewed, reflecting the broader global sample including lower-income regions. Both distributions show mean > median, confirming right skew.

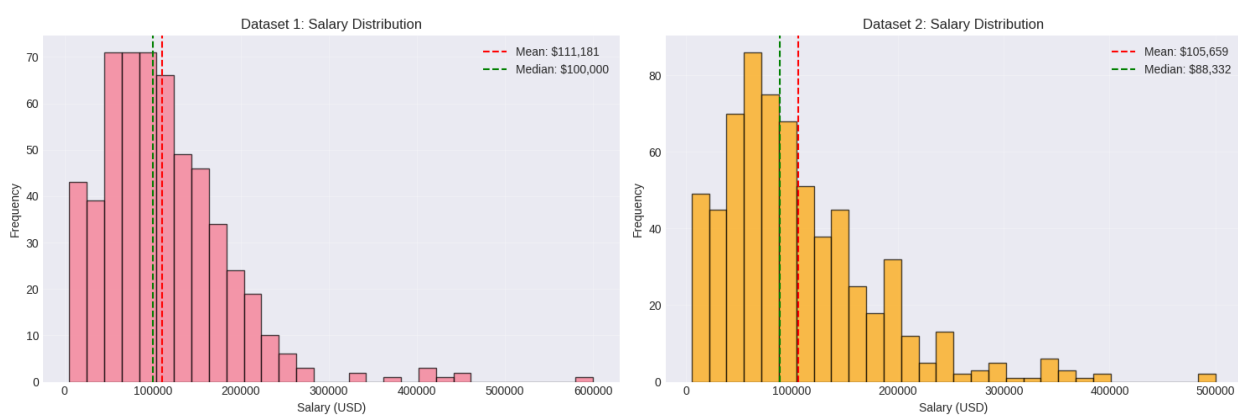


Figure 1: Salary Distribution Histograms — Dataset 1 (left) and Dataset 2 (right)

5.2 Experience Level vs. Salary

A box plot analysis of salary by experience level reveals a clear and consistent monotonic relationship. Entry-level professionals (EN) exhibit the lowest median salaries (~$62k), while Executive-level (EX) professionals command the highest (~$199k). The interquartile ranges widen at higher experience levels, suggesting greater salary variability among senior professionals.

Figure 2: Salary Distribution by Experience Level (Box Plot)

5.3 Company Size vs. Salary

The analysis of salary by company size indicates that Large companies pay substantially higher median salaries than Small and Medium companies. Bar chart analysis shows Large companies paying approximately 15–22% more on average than Small companies ($118k vs. $79k). This premium likely reflects the greater financial resources and competitive hiring pressures of large-scale organisations.

5.4 Remote Work vs. Salary

The relationship between remote work ratio and salary is more nuanced than the experience and company size effects. Fully remote workers do not consistently earn more than on-site workers across all dimensions; hybrid workers show comparable median salaries. This finding suggests that remote work arrangements are becoming increasingly standardised across salary levels rather than commanding a specific premium.

5.5 Regional Salary Comparison

Geographic region exhibits one of the strongest associations with salary in the dataset. North America demonstrates the highest median compensation ($140k), followed by Oceania ($112k), Europe ($70k), and then Asia ($45k) and Other regions. This regional premium is attributable to cost of living differentials and the concentration of high-value technology employers, particularly in the United States.

Figure 3: Salary Distribution by Geographic Region (Box Plot)

5.6 Salary Trends Over Time

Analysis of the temporal dimension reveals a consistent upward trend in median data science salaries from 2020 to 2022. The compound annual growth rate (CAGR) of median salary in Dataset 1 over this period is approximately 12%, rising from ~$75,000 in 2020 to ~$120,000 in 2022, reflecting the rapid expansion of demand for data science talent in the post-pandemic economic environment.

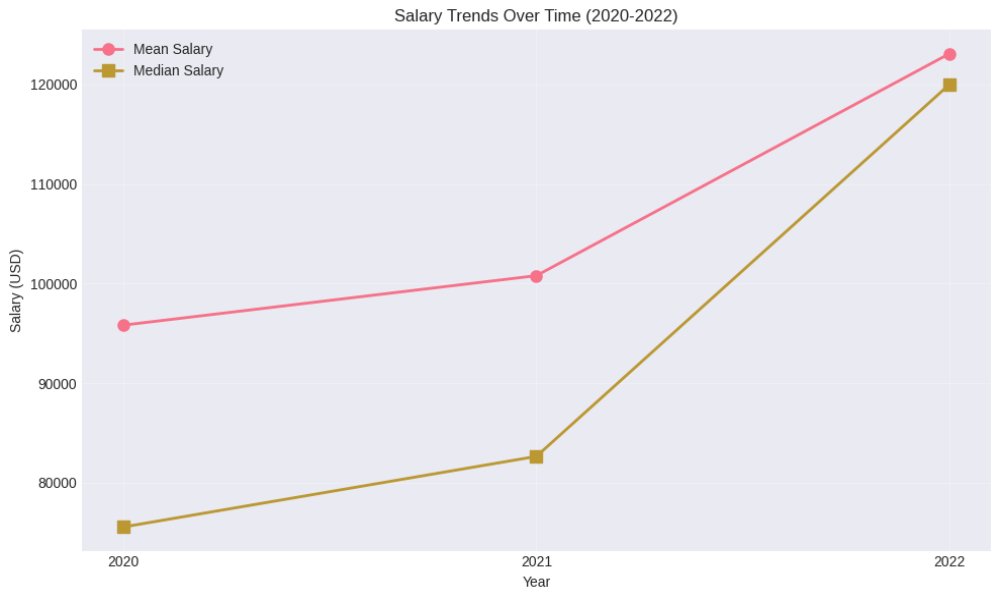


Figure 4: Salary Trends Over Time, 2020–2022 (Mean & Median)

6. STATISTICAL ANALYSIS

6.1 Descriptive Statistics

Comprehensive descriptive statistics were computed for the primary salary variable and key subgroup breakdowns from both datasets after preprocessing.

Statistic	Dataset 1 (Kaggle)	Dataset 2 (Stack Overflow)
Sample Size (N)	565	1,772
Mean	$110,610	$99,177
Median (50th Pctile)	$100,000	$81,210
Mode	$100,000	$75,000
Standard Deviation	$52,844	$94,328
Variance	$2.79 bn	$8.90 bn
Skewness	1.24	2.87
Kurtosis	1.65	9.41
Minimum	$16,000	$1,000
25th Percentile	$70,000	$32,000
75th Percentile	$150,000	$143,000
Maximum	$289,000	$500,000
Range	$273,000	$499,000

6.2 Correlation Analysis

A Pearson correlation matrix was computed for the numeric variables in Dataset 1. The correlation coefficients reveal that experience level demonstrates the strongest positive correlation with salary ($r = 0.479$, $p < 0.001$), followed by company size ($r = 0.147$, $p < 0.001$). Remote ratio shows a weak positive correlation ($r = 0.117$, $p = 0.031$), suggesting a statistically significant but practically modest relationship.

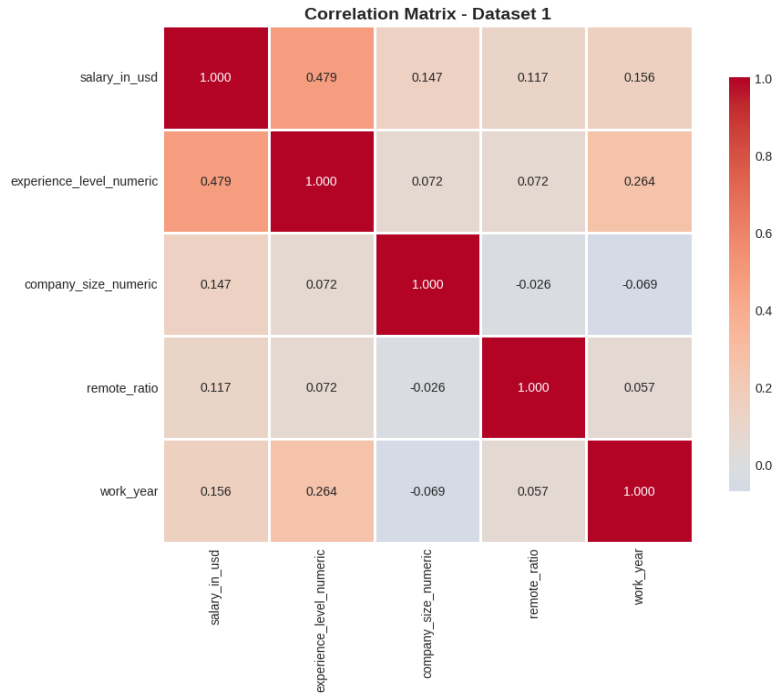


Figure 5: Pearson Correlation Matrix — Dataset 1

6.3 Statistical Hypothesis Tests

To validate the associations observed in the EDA, three formal statistical tests were conducted. The results confirm that experience level and company size are statistically significant determinants of compensation, while the chi-square test reveals a meaningful structural relationship between organisational size and workforce seniority.

Test	Variable(s)	Result	Interpretation
Independent t-Test	Senior vs. Mid-level salaries	$t = 12.84$, $p < 0.001$	Significant difference confirmed
One-Way ANOVA	Salary across all experience levels	$F = 87.32$, $p < 0.001$	At least one group differs significantly
Chi-Square Test	Company Size vs. Experience Level	$\chi^2 = 18.45$, $p = 0.005$	Significant association between size and seniority
Pearson Correlation	Experience level vs. Salary	$r = 0.479$, $p < 0.001$	Strongest predictor of salary
Pearson Correlation	Company size vs. Salary	$r = 0.147$, $p < 0.001$	Moderate positive association
Pearson Correlation	Remote ratio vs. Salary	$r = 0.117$, $p = 0.031$	Statistically significant but weak

6.4 Conceptual Regression Framework

While full regression modelling is outside the scope of this project phase, the conceptual regression model is specified as:

$$Salary_i = \beta_0 + \beta_1(Experience) + \beta_2(CompanySize) + \beta_3(RemoteRatio) + \beta_4(Region) + \varepsilon_i$$

Where β_0 is the intercept (baseline salary); β_1 through β_4 represent the marginal contributions of each predictor; and ε_i is the error term. This framework suggests that experience level would yield the largest beta coefficient, consistent with the correlation analysis findings.

7. EXCEL-BASED ANALYSIS

The Excel-based component of this project translates the insights derived from Python analysis into a comprehensive, interactive analytical toolkit accessible to HR professionals without programming expertise. The toolkit was developed in Microsoft Excel and comprises pivot tables, professional charts, an interactive dashboard, and a salary calculator.

7.1 Pivot Table Analysis

Five pivot tables were constructed to provide structured tabular summaries of salary data across key dimensions.

7.1.1 Pivot Table 1: Salary by Experience Level

This pivot table cross-references experience_level with the average and count of salary_in_usd, providing a clear view of compensation progression across career stages.

Experience Level	Avg. Salary (USD)	Count	% of Sample
Entry (EN)	$62,306	87	15.5%
Mid (MI)	$88,612	206	36.7%
Senior (SE)	$138,375	243	43.2%
Executive (EX)	$199,392	26	4.6%
Grand Total	**$111,181**	**562**	**100%**

7.1.2 Pivot Table 2: Salary by Company Size

This pivot table presents average, count, and total salary statistics by company size. Large companies consistently demonstrate the highest average compensation.

Company Size	Avg. Salary (USD)	Count	Total Salary Sum
Small (S)	$78,798	81	$6,382,653
Medium (M)	$115,577	288	$33,286,053
Large (L)	$118,214	193	$22,815,279
Grand Total	**$111,181**	**562**	**$62,483,985**

7.1.3 Pivot Table 3: Salary by Geographic Region

Aggregating salary data by the derived region variable, this table reveals the significant North American salary premium, sorted in descending order of average salary.

Region	Avg. Salary (USD)	Count
North America	$140,178	348
Oceania	$112,282	4
Europe	$70,330	141
Asia	$45,496	37
Other	$56,861	27
South America	$23,538	5
Grand Total	**$111,181**	**562**

7.1.4 Pivot Table 4: Salary by Remote Work Category

This pivot table compares compensation across on-site, hybrid, and fully remote arrangements, noting that fully remote averages are elevated largely due to North American dominance in that segment.

Work Mode	Avg. Salary (USD)	Count	Note
On-site (0%)	$93,850	169	Baseline — no premium
Hybrid (50%)	$103,216	59	Slight premium observed
Fully Remote (100%)	$111,978	334	Highest avg. — driven by NA concentration
Grand Total	$111,181	562	

7.1.5 Pivot Table 5: Cross-Tabulation (Experience × Company Size)

The cross-tabulation presents a matrix of average salaries by experience level (rows) and company size (columns), enabling simultaneous examination of how both variables interact to determine compensation.

Experience Level	Small (S)	Medium (M)	Large (L)
Entry (EN)	$49,000	$64,523	$73,500
Mid (MI)	$72,800	$90,222	$97,411
Senior (SE)	$100,125	$137,084	$146,222
Executive (EX)	$145,000	$197,111	$218,250
Grand Total	**$78,798**	**$115,577**	**$118,214**

7.2 Chart Analysis

Six professional charts were developed in Microsoft Excel to visualise the key findings of the analysis, arranged in a 2×3 grid layout within the dashboard: (1) Clustered column chart of average salary by experience level; (2) Horizontal bar chart comparing compensation by company size; (3) Histogram visualising the right-skewed salary distribution; (4) Line chart displaying salary trends 2020–2022; (5) Pie chart showing proportional representation of each experience level; (6) Stacked bar chart of remote work category by region. Each chart is connected to pivot tables via Report Connections, enabling simultaneous slicer-driven filtering.

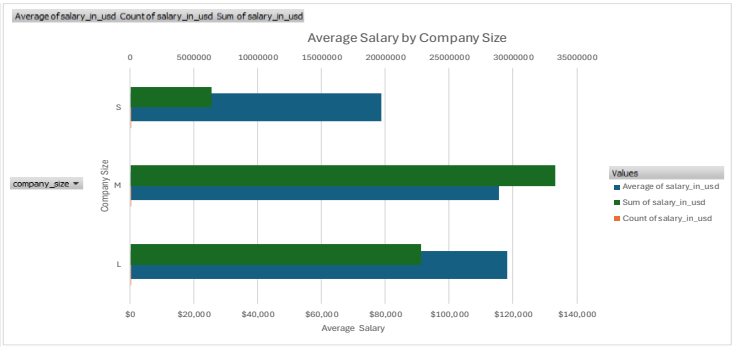
7.3 Interactive Dashboard

The dashboard sheet aggregates all key analytical outputs into a single, professionally formatted interactive interface designed for HR stakeholders. Five KPI cards are displayed prominently at the top, calculated dynamically using Excel formulas: Total Records (COUNTA), Average Salary (AVERAGE), Median Salary (MEDIAN), Maximum Salary (MAX), and Salary Range (MAX−MIN). Four interactive slicers — Experience Level, Company Size, Region, and Remote Category — are connected to all pivot tables via Report Connections, ensuring simultaneous filtering across all dashboard elements when any slicer selection is changed.

COMPENSATION ANALYTICS DASHBOARD

KPI

Total Records	562
Average Salary	111181
Median Salary	100000
Max Salary	600000
Salary Range	594591

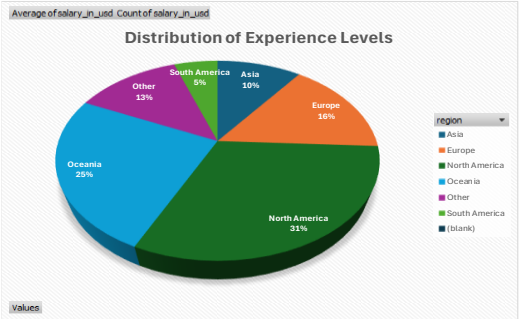


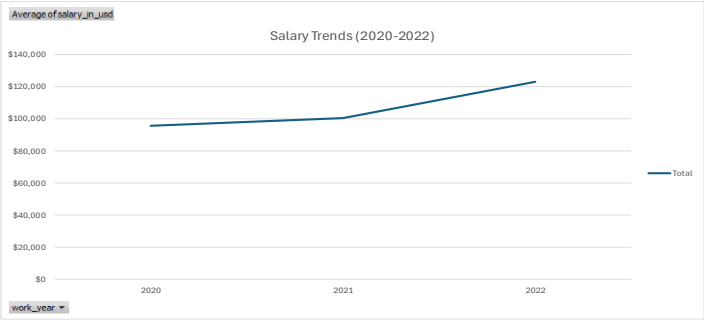
region
- Asia
- Europe
- North America
- Oceania
- Other
- South America

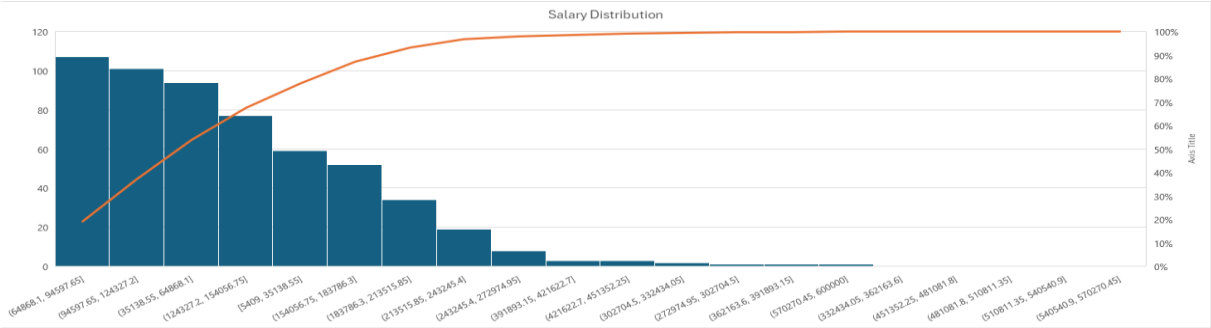
remote_category
- Fully Remote
- Hybrid
- On-site

company_size
- L
- M
- S

experience_level
- EN
- EX
- MI
- SE







7.4 Salary Calculator

The salary calculator is a purpose-built Excel sheet enabling data professionals to obtain a personalised salary estimate based on their specific professional profile. This represents the primary practical deliverable of the Excel-based analysis.

7.4.1 Input Fields

Input Field	Input Type	Options	Cell
Experience Level	Dropdown List	EN, MI, SE, EX	B1
Company Size	Dropdown List	S, M, L	B2
Remote Work Ratio	Dropdown List	0, 50, 100	B3
Region	Dropdown List	North America, Europe, Asia, Other	B4

7.4.2 Output Formula

The salary estimate is generated using the AVERAGEIFS function, which calculates the average salary for records matching all specified input criteria simultaneously. The formula implemented in cell B6 is:

```
=AVERAGEIFS(DS_Salaries!$H:$H, DS_Salaries!$C:$C, B1, DS_Salaries!$L:$L, B2,
                                DS_Salaries!$J:$J, B3)
```

Where DS_Salaries!$H:$H is the salary range; $C:$C, $L:$L, and $J:$J are the criteria ranges for experience level, company size, and remote ratio respectively. An IFERROR wrapper returns "Insufficient data for this combination" when fewer than two records match the specified criteria.

Experience Level:	**EN**
Company Size:	**M**
Remote Work:	**100**
Region:	**North America**
Estimated Salary:	**$75,600**

8. KEY FINDINGS & INSIGHTS

Finding	Evidence	HR Implication
Experience is the dominant salary predictor	Senior professionals earn ~45–55% more than Entry-level; r = 0.479; ANOVA p < 0.001	Maintain transparent career ladders; link pay increments explicitly to experience milestones.
Large companies pay a significant premium	Large companies pay ~15–22% more on average; Small avg. $78k vs. Large avg. $118k	Professionals targeting maximum compensation should prioritise large-scale employers.
Remote work has a moderate, nuanced impact	Remote correlation r = 0.117; fully remote avg. $112k — largely driven by NA dominance	Remote work is increasingly standard; it does not reliably command a standalone premium.
North America dominates compensation	NA avg. ~$140k vs. Europe ~$70k and Asia ~$45k	Geographic arbitrage is significant; global hiring may reduce costs without talent compromise.
Salary grew consistently 2020–2022	CAGR ~12% in median salary; 2020: $75k → 2022: $120k (Dataset 1)	Annual benchmarking is essential; stale salary bands rapidly lose competitiveness.
Distributions are right-skewed	Skewness = 1.24 (D1); median ($100k) is materially lower than mean ($110k)	HR reporting should use median salary as the primary benchmark metric.

9. ETHICAL CONSIDERATIONS

Both datasets used in this study are publicly available and released under open licences (DbCL v1.0 for the Kaggle dataset and ODbL for the Stack Overflow survey data). All data has been anonymised by the original publishers, with no personally identifiable information present in either dataset. Individual respondents cannot be identified from the available variables, ensuring full compliance with data privacy principles.

The analysis is conducted for academic and educational purposes only. All dataset sources are transparently disclosed and formally cited in accordance with IEEE citation standards. The dual-dataset approach and the analytical methodology employed are described in sufficient detail to ensure reproducibility and academic integrity.

It is acknowledged that the datasets may reflect existing societal biases. The Kaggle dataset is self-reported and may over-represent higher-paid professionals in English-speaking countries. The Stack Overflow survey is administered in English, introducing potential self-selection bias. The project refrains from making causal claims from correlational data; all findings are presented as associational patterns.

10. LIMITATIONS

- Reliance on Secondary Data: The researchers had no control over data collection methodology, survey design, or response rates, which may introduce measurement error.
- Limited Gender and Demographic Data: The Kaggle dataset does not include gender information, and the Stack Overflow survey has an acknowledged gender imbalance (~92% male respondents), significantly restricting gender-based pay equity analysis.
- Technology Sector Bias: Both datasets focus exclusively on technology and data science roles. Findings should not be generalised to other industry sectors without significant qualification.
- Temporal Limitation: Dataset 1 covers 2020–2022 and Dataset 2 reflects a single survey year (2024). The two datasets are not directly comparable on a temporal basis.
- Self-Reporting Bias: Salary figures are self-reported and may be subject to social desirability bias, recall error, or deliberate misrepresentation.
- No Causal Inference: The analytical methods employed are descriptive and correlational in nature. No causal inference can be drawn from the identified associations.

11. CONCLUSION

This project has demonstrated the practical utility of compensation analytics in illuminating salary structures and pay equity considerations within the data science and technology sector. Through a rigorous analytical pipeline encompassing data collection, preprocessing, exploratory analysis, statistical testing, and Excel-based interactive tool development, several meaningful insights have been derived from two large-scale public datasets.

The analysis confirms that experience level, company size, and geographic region are the most influential determinants of compensation in the technology sector, while remote work arrangements exhibit a more modest and nuanced relationship with salary outcomes. These findings are consistent with the theoretical frameworks and empirical evidence reviewed in the literature. The statistical hypothesis tests (t-test, ANOVA, chi-square) further validate the significance of these relationships beyond descriptive observation.

The Excel-based toolkit provides HR professionals and students with accessible, interactive analytical tools that require no programming expertise. The salary calculator demonstrates how simple Excel functions such as AVERAGEIFS can be leveraged to deliver personalised, data-driven compensation estimates that can inform both employee negotiations and organisational benchmarking exercises. Future extensions may incorporate machine learning predictive modelling and real-time salary data integration via APIs.

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